WORKED SOLUTION

STEP III 1995, Question 1

Pure • Simultaneous Linear Equations

STEP III 1995, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q1

The key idea

The second and third equations have the same y and z terms, so subtracting one from the other removes two unknowns at once and leaves a single equation for x. Most of the question is about the values of a and b for which that step, or a later division, breaks down. The determinant of the coefficients tells us in advance where those values are, so we compute it first and then treat each exceptional case on its own.

Solution

Label the equations

$$a^2x + ay + z = a^2 \tag{1}$$
$$ax + y + bz = 1 \tag{2}$$
$$a^2bx + y + bz = b. \tag{3}$$

Where can things go wrong?

Expanding the determinant of the coefficient matrix along the first row,

$$\begin{vmatrix} a^2 & a & 1 \\ a & 1 & b \\ a^2b & 1 & b \end{vmatrix} = a^2(b-b) - a(ab - a^2b^2) + (a - a^2b)$$
$$= a^3b^2 - 2a^2b + a = a(ab-1)^2.$$

So there is exactly one solution unless $a = 0$ or $ab = 1$. In those cases there are either no solutions or infinitely many, and we have to look at the equations themselves to decide which.

The general case: $a \neq 0$ and $ab \neq 1$

Subtracting (2) from (3) gets rid of y and z:

$$(a^2b - a)x = b - 1, \qquad \text{that is,} \qquad a(ab-1)\,x = b - 1.$$

Since $a \neq 0$ and $ab \neq 1$ we may divide, giving

$$x = \frac{1-b}{a(1-ab)}.$$

Next, (1) gives $z = a^2 - a^2x - ay$. Substituting this into (2):

$$ax + y + a^2b - a^2bx - aby = 1 \quad \Longrightarrow \quad (1-ab)\,y = (1-a^2b) - ax(1-ab).$$

Dividing by $1 - ab \neq 0$ and using $ax = \dfrac{1-b}{1-ab}$,

$$y = \frac{1-a^2b}{1-ab} - \frac{1-b}{1-ab} = \frac{b-a^2b}{1-ab} = \frac{b(1-a^2)}{1-ab}.$$

Finally,

$$\begin{aligned} z = a^2 - a(ax) - ay &= a^2 - \frac{a(1-b)}{1-ab} - \frac{ab(1-a^2)}{1-ab} \\ &= \frac{a^2 - a^3b - a + ab - ab + a^3b}{1-ab} = \frac{a(a-1)}{1-ab}. \end{aligned}$$

So when $a \neq 0$ and $ab \neq 1$ the unique solution is

$$\boxed{x = \frac{1-b}{a(1-ab)}, \qquad y = \frac{b(1-a^2)}{1-ab}, \qquad z = \frac{a(a-1)}{1-ab}.}$$

The case $a = 0$

The equations become

$$z = 0, \qquad y + bz = 1, \qquad y + bz = b,$$

so $z = 0$, $y = 1$, and the last equation then says $1 = b$. Hence:

- if $a = 0$ and $b \neq 1$, there are **no solutions**;
- if $a = 0$ and $b = 1$, the solutions are $x = \lambda$, $y = 1$, $z = 0$ for any real λ: a line of solutions, since x no longer appears in any equation.

The case $ab = 1$ (so $a \neq 0$)

Ask yourself When $ab = 1$, the equation $a(ab-1)x = b - 1$ has left-hand side 0. What does that tell us?

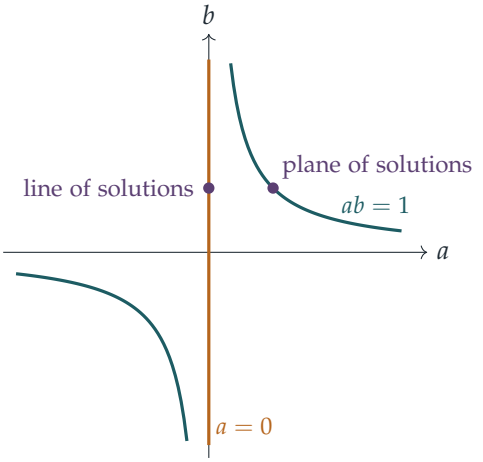
Subtracting (2) from (3) is still valid; it now reads $0 = b - 1$. So if $b \neq 1$ the equations are inconsistent. With $ab = 1$, the condition $b = 1$ forces $a = 1$. Hence:

- if $ab = 1$ and $a \neq 1$, there are **no solutions**;
- if $a = b = 1$, all three equations become $x + y + z = 1$, so the solutions are $x = \lambda$, $y = \mu$, $z = 1 - \lambda - \mu$ for any real λ, μ: a whole plane of solutions.

This fits with the squared factor $(ab - 1)^2$ in the determinant. At $a = b = 1$ the matrix drops all the way to rank 1, so there are two free parameters, while at $a = 0$, $b = 1$ it has rank 2 and there is only one.

Summary

The picture below shows the (a, b)-plane. Away from the line $a = 0$ and the hyperbola $ab = 1$ there is a unique solution. On them there are no solutions, except at the two marked points.



- $a \neq 0$, $ab \neq 1$: the unique solution boxed above.
- $a = 0$, $b = 1$: $(x, y, z) = (\lambda, 1, 0)$, $\lambda \in \mathbb{R}$.
- $a = b = 1$: $(x, y, z) = (\lambda, \mu, 1 - \lambda - \mu)$, $\lambda, \mu \in \mathbb{R}$.
- $a = 0$ with $b \neq 1$, or $ab = 1$ with $a \neq 1$: no solutions.

Checks

Take $a = 2$, $b = 3$. The formulae give $x = \frac{1}{5}$, $y = \frac{9}{5}$, $z = -\frac{2}{5}$, and indeed $4 \cdot \frac{1}{5} + 2 \cdot \frac{9}{5} - \frac{2}{5} = 4 = a^2$, $\frac{2}{5} + \frac{9}{5} - \frac{6}{5} = 1$ and $\frac{12}{5} + \frac{9}{5} - \frac{6}{5} = 3 = b$. Another quick test is $b = 1$, $a \neq 0, 1$: the formulae give $x = 0$, $y = 1 + a$, $z = -a$, and these satisfy all three equations. We also checked the determinant and the general formulae with exact rational arithmetic for 2000 random pairs (a, b).

Common slips

- Dividing by $a(ab-1)$ without first setting aside $a=0$ and $ab=1$. "For all possible values" means the special cases are the point of the question.
- Saying "determinant zero, so no solutions". A zero determinant only rules out a *unique* solution; here two of the special parameter values give infinitely many.
- Missing that $ab=1$ and $b=1$ together force $a=1$, so there is just one special point on the hyperbola.

Beyond STEP

This is a small example of how the rank of a matrix controls a linear system. The system is consistent exactly when the coefficient matrix and the augmented matrix have the same rank, and then the solutions form a set of dimension $3-\text{rank}$. That is why we got a single point, a line or a plane. Gaussian elimination makes all of this systematic, and Strang's *Introduction to Linear Algebra* develops it with many worked examples.

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